

Q1. Data types & ML algorithms

(a)

- Numerical: Age, Blood Pressure
- Categorical: Disease type, Treatment success (Yes/No)

(b) Using categorical data as numbers can mislead models because numeric values imply order/magnitude (e.g., disease 1 < disease 2), which may not exist.

(c)

- Handles categorical well: **Decision Trees, Random Forests**
 - Fits numerical better: **Linear Regression, SVM, KNN**
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Q2. Central tendency & spread

(a) Use **median** — mean is skewed by outliers like the CEO's 1 crore salary.

(b) Variance measures how spread out data is from the mean.

Example: Salaries [50k, 52k, 48k] → low variance; [50k, 1cr, 52k] → high variance.

(c) Standard deviation is in **same units as data**, easier to interpret than variance which is squared units.

Q3. Overfitting & underfitting

(a) Memorizing past papers = model fits training data too closely → fails on new data (overfitting).

(b) Strategies:

1. Use more data
2. Regularization (or simplify model)

(c) Analogy: Underfitting = student studies too little → fails; Overfitting = studies only old questions → fails too. Both give poor performance.

Q4. Missing values & outliers

(a) Handle missing income:

1. Impute (mean/median)
2. Remove rows with missing values

(b) Outlier: 100× higher house price → can skew model, giving wrong predictions.

(c) Systematic missing values are worse; they bias the model, unlike random missing which may cancel out.

Q5. Data visualization

(a) Long left tail → most students scored high, few scored very low (negatively skewed).

(b) Dots above whisker → outliers.

(c) Scatter plots show patterns/trends → helps guess relationships before modeling.

Q6. Features & targets

(a) Features: area, bedrooms, location, age; Target: house price

(b) Strong correlation → model predicts target more accurately.

(c) Multiple targets possible: Predicting house **price and rent** together.

Q7. ML types

(a)

- Group songs by similarity → **Unsupervised**
- Predict rainfall → **Supervised**
- Teach robot to balance → **Reinforcement Learning**

(b) Supervised is common → labeled data is easier to evaluate and apply.

(c) Unsupervised benefit → finds hidden patterns/clusters without labels.

Q8. Data bias

- (a) More male applicants → **sampling bias / representation bias**
 - (b) Model may favor majority group → unfair predictions.
 - (c) Reduce bias → oversample minority group or reweight data.
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Q9. Model complexity vs data size

- (a) Small dataset → use **simple model** (less chance of overfitting).
- (b) Too many parameters → model memorizes data, poor generalization.
- (c) Simpler models → generalize better, avoid overfitting on small datasets.